

```

1. #include<iostream>

using namespace std;

int main(){

    string words[] = {"abc","car","ada","racecar","cool"};

    for(int i=0; i < 5 ; i++){

        string s = words[i];

        string rev = "";

        for(int j = s.length()-1 ; j>= 0; j--){

            rev = rev + s[j];

        }

        if (s == rev){

            cout << s <<" is palindrome!!";

            break;

        }

    }

    return 0;

}

```

```
ada is palindrome!!
```

```
-----
Process exited after 0.2872 seconds with return value 0
Press any key to continue . . .
```

```

2.#include<iostream>

using namespace std;

int main(){

    int nums1[] = {2 , 3 , 2};

    int nums2[] = {1 , 2};


    int answer1 = 0 , answer2=0;


    for (int i = 0 ; i < 3 ; i++){

        for(int j = 0 ; j < 2; j++){

            if(nums1[i] ==nums2[j]){

                answer1++;

                break;

            }

        }

    }

    for (int i = 0; i < 2; i++){

        for (int j = 0 ; j < 3 ; j++){

            if (nums2[i] == nums1[j]){

                answer2++;

                break;

            }

        }

    }

    cout << "["<<answer1<<","<<answer2<<"]";

    return 0;

}

```

```
C:\Users\User\OneDrive\Docu  ×  +  ∨  
[2,1]  
-----  
Process exited after 0.2724 seconds with return value 0  
Press any key to continue . . . |
```

```
3. #include<iostream>  
  
#include<set>  
  
using namespace std;  
  
int main(){  
    int n;  
    cin >> n;  
  
    int nums[10];  
    for(int i = 0 ; i < n ; i++){  
        cin >> nums[i];  
    }  
    long long sum = 0;  
    for(int i = 0; i < n ; i++){  
        set<int> s;  
        for(int j = 1 ; j < n ; j++){  
            s.insert(nums[j]);  
  
            int distinct = s.size();  
            sum = sum + (distinct * distinct);  
        }  
    }  
    cout << sum << endl;  
    return 0;
```

```

3
1
2
1
15
-----
Process exited after 4.568 seconds with return value 0
Press any key to continue . . . |
}

```

4. #include <iostream>

using namespace std;

int main() {

int n, k;

int count = 0;

int mul;

cin >> n;

int nums[100];

for (int i = 0; i < n; i++) {

cin >> nums[i];

}

cin >> k;

for (int i = 0; i < n; i++) {

for (int j = 0; j < n; j++) {

if (i < j) {

if (nums[i] == nums[j]) {

```

        mul = i * j;

        if (mul % k == 0) {
            count++;
        }
    }
}

cout << count;

return 0;
}

```

```

7
3
2
3
3
2
1
3
4
count is:5
-----
Process exited after 11 seconds with return value 0
Press any key to continue . . .

```

```

5. include<iostream>

using namespace std;

int main(){

    int n ;

    cin >> n;

    int nums[10];

    for(int i = 0 ; i < n ; i++){

        cin >> nums[i];
    }
}

```

```

    }

    int max = nums[0];
    for( int i = 1 ; i < n ; i++){
        if(max = nums[i]){
            max = nums[i];
        }
    }

    cout << max;

    return 0 ;
}

```

```

5
1
2
3
4
5
5
-----
Process exited after 4.739 seconds with return value 0
Press any key to continue . . . |

```

6. #include <iostream>

#include <algorithm>

using namespace std;

int main() {

int n;

cin >> n;

if (n == 0) {

cout << "List is empty";

return 0;

}

int arr[100];

```

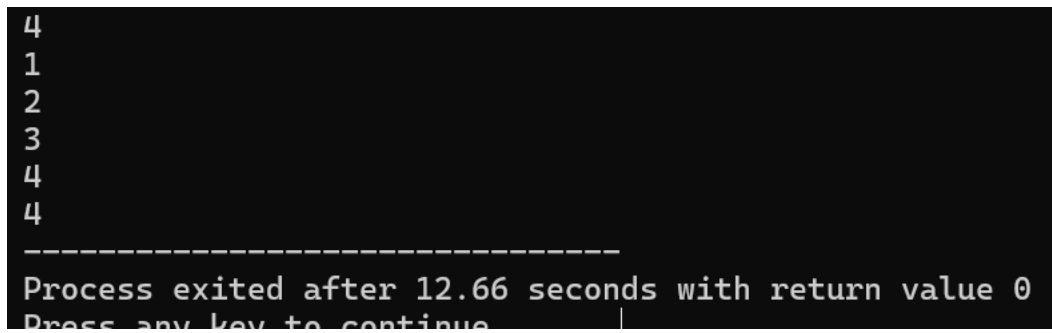
for (int i = 0; i < n; i++) {
    cin >> arr[i];
}

// Sort the array
sort(arr, arr + n);

// Maximum element is the last element
cout << arr[n - 1];

return 0;
}

```



```

4
1
2
3
4
4
-----
Process exited after 12.66 seconds with return value 0
Press any key to continue

```

7. #include <iostream>

using namespace std;

```

int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;

    int arr[100];
    cout << "Enter " << n << " numbers: ";
    for (int i = 0; i < n; i++) {
        cin >> arr[i];
    }
}

```

```
}
```

```
int uniq[100];
```

```
int count = 0;
```

```
for (int i = 0; i < n; i++) {
```

```
    bool found = false;
```

```
    for (int j = 0; j < count; j++) {
```

```
        if (arr[i] == uniq[j]) {
```

```
            found = true;
```

```
            break;
```

```
        }
```

```
    }
```

```
    if (!found) {
```

```
        uniq[count] = arr[i];
```

```
        count++;
```

```
    }
```

```
}
```

```
cout << "Unique elements: ";
```

```
for (int i = 0; i < count; i++) {
```

```
    cout << uniq[i] << " ";
```

```
}
```

```
cout << endl;
```

```
return 0;
```

```
}
```

```
Enter n: 6
Enter 6 numbers: 1
2
3
4
5
2
Unique elements: 1 2 3 4 5

-----
Process exited after 8.585 seconds with return value 0
Press any key to continue . . .
```

```
8. #include <iostream>
```

```
using namespace std;
```

```
int main() {
```

```
    int n;
```

```
    cout << "Enter n: ";
```

```
    cin >> n;
```

```
    int arr[100];
```

```
    cout << "Enter " << n << " numbers: ";
```

```
    for (int i = 0; i < n; i++) {
```

```
        cin >> arr[i];
```

```
    }
```

```
    for (int i = 0; i < n - 1; i++) {
```

```
        bool swapped = false;
```

```
        for (int j = 0; j < n - i - 1; j++) {
```

```
            if (arr[j] > arr[j + 1]) {
```

```
                int temp = arr[j];
```

```
                arr[j] = arr[j + 1];
```

```
                arr[j + 1] = temp;
```

```
                swapped = true;
```

```
            }
```

```

    }

    if (!swapped) break; // already sorted, stop early
}

cout << "Sorted array: ";
for (int i = 0; i < n; i++) {
    cout << arr[i] << " ";
}
cout << endl;

return 0;
}

```

```

Enter n: 6
Enter 6 numbers: 15
73
45
34
23
56
Sorted array: 15 23 34 45 56 73

-----
Process exited after 17.98 seconds with return value 0
Press any key to continue . . . |

```

9. #include <iostream>

using namespace std;

```

int main() {
    int n, key;

    cout << "Enter n: ";
    cin >> n;

    int arr[100];

    cout << "Enter " << n << " numbers: ";
    for (int i = 0; i < n; i++) {

```

```

    cin >> arr[i];
}
cout << "Enter key: ";
cin >> key;

// Step 1: sort (bubble sort)
for (int i = 0; i < n - 1; i++) {
    for (int j = 0; j < n - i - 1; j++) {
        if (arr[j] > arr[j + 1]) {
            int temp = arr[j];
            arr[j] = arr[j + 1];
            arr[j + 1] = temp;
        }
    }
}

// Step 2: binary search
int low = 0, high = n - 1;
int pos = -1;

while (low <= high) {
    int mid = (low + high) / 2;
    if (arr[mid] == key) {
        pos = mid + 1;    // 1-based position
        break;
    } else if (arr[mid] < key) {
        low = mid + 1;    // search right half
    } else {
        high = mid - 1;   // search left half
    }
}

```

```

if (pos != -1)

    cout << "Element " << key << " is found at position " << pos << endl;

else

    cout << "Element " << key << " is not found" << endl;

return 0;

}

```

```

Enter n: 7
Enter 7 numbers: 4
6
7

4
2
5
5
Enter key: 5
Element 5 is found at position 4

-----
Process exited after 15.34 seconds with return value 0
Press any key to continue . . . |

```

10. #include <iostream>

using namespace std;

```
int main() {
```

```
    int n, a[100], temp[100];
```

```
    cin >> n;
```

```
    for (int i = 0; i < n; i++)
```

```
        cin >> a[i];
```

```
    // Merge sort
```

```
    for (int size = 1; size < n; size = size * 2) {
```

```

for (int left = 0; left < n - 1; left += 2 * size) {

    int mid = left + size - 1;
    int right = left + 2 * size - 1;

    if (mid >= n)
        continue;

    if (right >= n)
        right = n - 1;

    int i = left, j = mid + 1, k = left;

    while (i <= mid && j <= right) {
        if (a[i] < a[j])
            temp[k++] = a[i++];
        else
            temp[k++] = a[j++];
    }

    while (i <= mid)
        temp[k++] = a[i++];

    while (j <= right)
        temp[k++] = a[j++];

    for (i = left; i <= right; i++)
        a[i] = temp[i];
    }
}

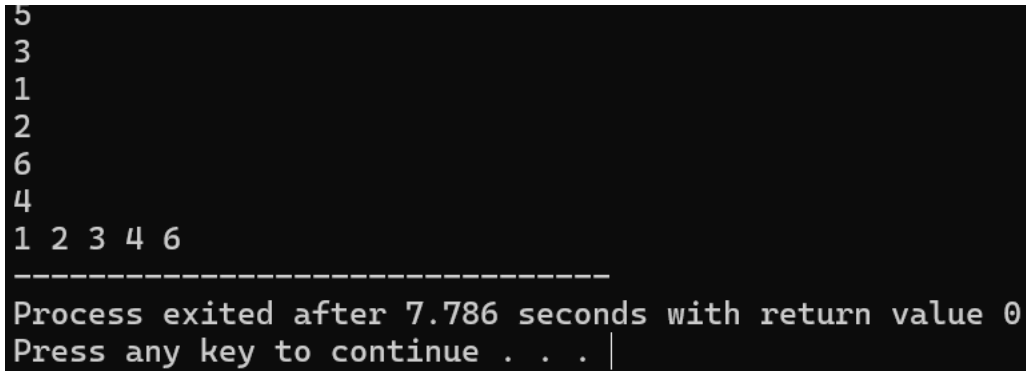
```

```

    for (int i = 0; i < n; i++)
        cout << a[i] << " ";

    return 0;
}

```



```

5
3
1
2
6
4
1 2 3 4 6
-----
Process exited after 7.786 seconds with return value 0
Press any key to continue . . . |

```

11. #include <iostream>

using namespace std;

```
int main() {
```

```
    int m, n, N, startI, startJ;
```

```
    cin >> m >> n >> N >> startI >> startJ;
```

```
    int dp[20][20] = {0};
```

```
    dp[startI][startJ] = 1;
```

```
    int answer = 0;
```

```
    for (int step = 1; step <= N; step++) {
```

```
        int next[20][20] = {0};
```

```
        for (int i = 0; i < m; i++) {
```

```
            for (int j = 0; j < n; j++) {
```

```

    if (dp[i][j] == 0)
        continue;

    // Up
    if (i == 0)
        answer += dp[i][j];
    else
        next[i - 1][j] += dp[i][j];

    // Down
    if (i == m - 1)
        answer += dp[i][j];
    else
        next[i + 1][j] += dp[i][j];

    // Left
    if (j == 0)
        answer += dp[i][j];
    else
        next[i][j - 1] += dp[i][j];

    // Right
    if (j == n - 1)
        answer += dp[i][j];
    else
        next[i][j + 1] += dp[i][j];
}
}

// Move next positions into dp
for (int i = 0; i < m; i++)

```

```

        for (int j = 0; j < n; j++)
            dp[i][j] = next[i][j];
    }

    cout << answer;

    return 0;
}

```

```

2
2
2
0
0
6
-----
Process exited after 4.528 seconds with return value 0
Press any key to continue . . . |

```

12. #include <iostream>

using namespace std;

```
int main() {
```

```
    int n;
```

```
    cin >> n;
```

```
    int a[100];
```

```
    for (int i = 0; i < n; i++)
```

```
        cin >> a[i];
```

```
    if (n == 1) {
```

```
        cout << a[0];
```

```
        return 0;
```

```
}
```

```
// Case 1: Don't rob the last house
```

```
int prev2 = 0, prev1 = 0;
```

```
for (int i = 0; i < n - 1; i++) {
```

```
    int current = max(prev1, prev2 + a[i]);
```

```
    prev2 = prev1;
```

```
    prev1 = current;
```

```
}
```

```
int case1 = prev1;
```

```
// Case 2: Don't rob the first house
```

```
prev2 = 0;
```

```
prev1 = 0;
```

```
for (int i = 1; i < n; i++) {
```

```
    int current = max(prev1, prev2 + a[i]);
```

```
    prev2 = prev1;
```

```
    prev1 = current;
```

```
}
```

```
int case2 = prev1;
```

```
cout << max(case1, case2);
```

```
return 0;
```

```
4
2
3
5
3
7
-----
Process exited after 11.92 seconds with return value 0
Press any key to continue . . . |
```

13. #include <iostream>

using namespace std;

int main() {

int n;

cin >> n;

int a = 1;

int b = 2;

if (n == 1) {

cout << 1;

return 0;

}

if (n == 2) {

cout << 2;

return 0;

}

for (int i = 3; i <= n; i++) {

int c = a + b;

a = b;

b = c;

}

```

cout << b;

return 0;
}

```

```

4
5
-----
Process exited after 2.846 seconds with return value 0
Press any key to continue . . . |

```

14. #include <iostream>

using namespace std;

```

int main() {
    int m, n;
    cin >> m >> n;

    int a[100][100];

    // First row = 1
    for (int j = 0; j < n; j++)
        a[0][j] = 1;

    // First column = 1
    for (int i = 0; i < m; i++)
        a[i][0] = 1;

    // Fill remaining cells
    for (int i = 1; i < m; i++) {
        for (int j = 1; j < n; j++) {
            a[i][j] = a[i - 1][j] + a[i][j - 1];
        }
    }
}

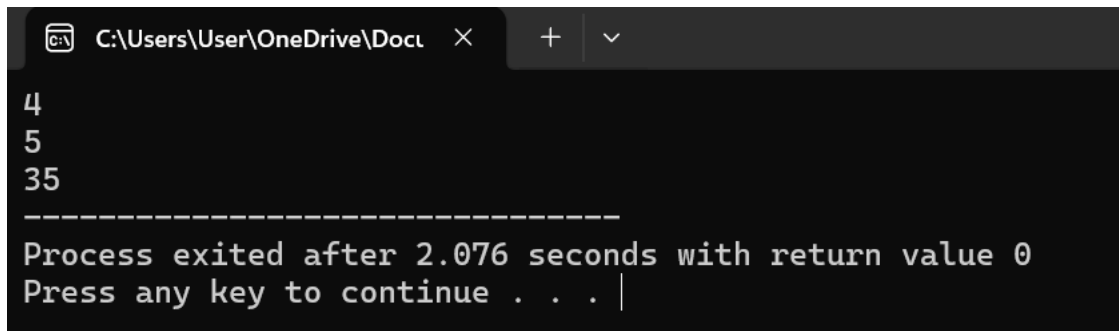
```

```
}
```

```
cout << a[m - 1][n - 1];
```

```
return 0;
```

```
}
```



```
C:\Users\User\OneDrive\Docu  ×  +  ∨  
4  
5  
35  
-----  
Process exited after 2.076 seconds with return value 0  
Press any key to continue . . . |
```

15.

```
using namespace std;
```

```
int main() {
```

```
    string s;
```

```
    cin >> s;
```

```
    int start = 0;
```

```
    for (int i = 1; i <= s.length(); i++) {
```

```
        if (i == s.length() || s[i] != s[i - 1]) {
```

```
            int length = i - start;
```

```
            if (length >= 3) {
```

```
                cout << "[" << start << "," << i - 1 << "]" ";
```

```
            }
```

```

        start = i;
    }
}

return 0;
}

```

```

abbxxxxzzy
[3,6]
-----
Process exited after 0.9507 seconds with return value 0
Press any key to continue . . . |

```

16. #include <iostream>

using namespace std;

```
int main() {
```

```
    int m, n;
```

```
    cin >> m >> n;
```

```
    int board[100][100];
```

```
    int next[100][100];
```

```
    // Input
```

```
    for (int i = 0; i < m; i++) {
```

```
        for (int j = 0; j < n; j++) {
```

```
            cin >> board[i][j];
```

```
        }
```

```
    }
```

```
    // Check every cell
```

```
    for (int i = 0; i < m; i++) {
```

```
        for (int j = 0; j < n; j++) {
```

```

int count = 0;

// Check 8 neighbours
for (int x = i - 1; x <= i + 1; x++) {
    for (int y = j - 1; y <= j + 1; y++) {

        if (x >= 0 && x < m &&
            y >= 0 && y < n &&
            !(x == i && y == j)) {

            count += board[x][y];
        }
    }
}

// Apply rules
if (board[i][j] == 1) {
    if (count == 2 || count == 3)
        next[i][j] = 1;
    else
        next[i][j] = 0;
}
else {
    if (count == 3)
        next[i][j] = 1;
    else
        next[i][j] = 0;
}
}
}

```

```
// Copy next state to board
for (int i = 0; i < m; i++) {
    for (int j = 0; j < n; j++) {
        board[i][j] = next[i][j];
    }
}

// Print result
for (int i = 0; i < m; i++) {
    for (int j = 0; j < n; j++) {
        cout << board[i][j] << " ";
    }
    cout << endl;
}

return 0;
}
```